RAGHAV GULATI

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ACADEMIC ACHIEVEMENTS

Year	Degree / Board	Institute	SGPA / Marks
2024-28	B.Tech CS with Applied Mathematics (Ongoing)	VIPS, New Delhi	**9.28**
2024	Class XII, Non-Med with CS / CBSE	Salwan Public School, New Delhi	85%
2022	Class X, CBSE	Salwan Public School, New Delhi	87.5%

TECHNICAL SKILLS

- **Languages:** Python, C, JavaScript, HTML, CSS, Bootstrap, Dart
- **AI :** FastAPI, React, Groq API, Ollama, Qdrant, fastembed, prompting
- **Frameworks & Tools:** Flutter, Android Studio, MySQL, Git/GitHub (active since 2020), Google Colab, Jupyter Notebook

SCHOLASTIC ACHIEVEMENTS

- **Ideathon Winner: Satakunta University, Finland: 1st place + LOR**; winning concept behind Shruti AI. [2022]
- **iCode Hackathon:** Ranked **98th** nationally **(top 1.3% of 8,000+)** & **178th** globally among **50,000+** participants, Python Advanced. [2023]
- **Delhi University CIC Cluster: Ranked 2** in DU CIC college in a 24-hour hackathon. [2026]
- **IIIT DELHI: Ranked 2** in IIITD in a 24-hour hackathon. [2026]
- **Shri Ram Global School Animation Competition: Rank 1** in storytelling via animation competition. [2024]
- **Cybersecurity Speaker:** 5+ workshops on online safety, phishing & data privacy at Salwan Public School. [2022-2024]

PROJECTS

NyayaSetu - *AI-Powered Indian Legal Research Assistant* *2026 - Current*

Stack: Python, FastAPI, React, Groq API, Ollama, Qdrant

Assembled a **full-stack legal AI platform** for Indian advocates and legal interns, integrating **RAG-based semantic search** over local legal document corpora using **Qdrant vector DB and fastembed (BAAI/bge-small-en-v1.5, 384-dim)** embeddings, paired with **live case scraping from Indian Kanoon**. Architected a **dual-LLM pipeline** routing queries through **Groq (LLaMA 3.1) with Ollama** fallback to ensure continuity during API exhaustion. Engineered an **AES-256 encrypted** query logging system and **JWT-based authentication** to meet legal confidentiality requirements. Implemented a **compliance alert scheduler** with 12-hour auto-refresh cycles and a **modular route architecture** spanning case search, document automation, legal aid, and workflow management.

Shruti AI - *Privacy-First Personal AI Companion* *2022 - Current*

stack: Python, Ollama, Phi-3 Mini, Local LLM Systems

Built a **fully offline AI companion** using Python and Ollama (Phi-3 Mini), eliminating cloud dependency and ensuring complete data privacy through local-only storage. Architected a **modular system of 6+ NLP modules** handling conversational planning, contextual memory management, and multilingual interaction across English, Hindi and Hinglish. Implemented **consent-based memory storage with Ebbinghaus decay algorithms** to simulate human-like memory reinforcement and forgetting over time. Engineered **conversation-state tracking and question-control logic** to maintain contextual continuity and avoid repetitive interactions, alongside emotional context detection and sarcasm uncertainty handling for conversational naturalness and safety.

TruthLens AI - *Misinformation Detection - Chrome Extension* *2025*

Stack: JavaScript · Chrome Manifest V3 · Content Scripts · Background Service Workers · Shadow DOM

Architected a **full-stack AI-powered browser extension** that analyses text and images to assess the likelihood of AI-generated or manipulated content using a **multi-signal ensemble approach**. Integrated a FastAPI backend combining **vision-language models (CLIP), frequency-domain analysis (FFT), and NLP-based text evaluation** through a confidence-aware fusion engine to deliver transparent, risk-based verdicts. Shipped a Chrome Extension (Manifest V3) featuring **full-page scanning, selective text/image analysis, snip-based detection, and real-time UI overlays** with explainable outputs. Prioritised **uncertainty-aware decision making** throughout - avoiding overconfident predictions and improving trustworthiness in AI detection workflows.

Hand Movement Tracker - *Real-time Touchless Gesture Control* *2025*

Stack: Python · OpenCV 4 MediaPipe Hands (21-landmark model) · Coordinate Mapping

Engineered a **real-time gesture control system** using OpenCV and MediaPipe's 21-landmark hand model, enabling touchless object manipulation through live webcam input with no additional hardware. Mapped **wrist angle to rotation delta and thumb-index spread to scale factor**, delivering intuitive two-axis control at **~30 FPS on a standard laptop**. Applied a **5-pixel jitter threshold debounce filter** that reduces false trigger rate by ~80% compared to raw coordinate tracking, significantly improving interaction stability. Added a **real-time landmark visualisation overlay** in OpenCV for live calibration and debug output.

Student Dashboard App - Cross-Platform Mobile App 2025

Stack: Flutter · Dart · Material Design · Named Dynamic Routing · Android & iOS

Shipped a **cross-platform student portal from a single Flutter/Dart codebase**, deploying to Android and iOS with zero platform-specific code. Designed a **named dynamic routing system** spanning four core modules — syllabus, fees, results and timetable — with seamless navigation and shared state across the app. Applied **stateless and stateful widget architecture** with Material Design theming and dynamic JSON data binding to simulate a live, data-driven portal experience. Structured the codebase with **clear separation of UI, routing and data layers** to maintain scalability as modules grow.

Python Web Scraper - *Configurable Automation CLI Tool* *2024*

Stack: Python · Requests · BeautifulSoup4 · CSV / TXT Export · Pagination Handling

Designed a **configurable CLI scraping tool** built for non-technical users, with **flag-based input (-keyword, -pages, -output)** requiring zero code modification to operate across different targets. Implemented **multi-page pagination handling and keyword filtering** with dual-format export to CSV and TXT for flexible downstream use. Hardened the pipeline with **HTTP session reuse, redirect handling and malformed HTML recovery** via BeautifulSoup4's lxml backend, ensuring reliability across inconsistent web sources. Structured as a **modular CSS-selector-targeted pipeline** making it straightforward to extend for new sites or data formats without touching core logic..

EXPERIENCE

Freelance Web Developer *2025*

Stack: HTML, CSS, JavaScript, Netlify

Developed and delivered a production-ready medical tourism website for a client by translating business requirements into a structured, user-friendly web application. Designed responsive multi-section layouts including service listings, informational pages, and contact flows, while refining code through debugging and UI improvements to ensure functionality and alignment with client expectations. Improved navigation and overall user experience across devices, and managed end-to-end deployment using Netlify with iterative enhancements based on client feedback.

Live: https://magnificent-dodol-a58fef.netlify.app/

Vivekananda Institute of Professional Studies - Entrepreneur Summit *2024-Current*

Contributed to the planning and execution of a large-scale Entrepreneur Summit by managing ticketing, sales, and financial operations. Led ticket distribution for **500+** attendees while overseeing transactions exceeding **₹1.5 lakh** with complete accuracy. Coordinated sponsorship outreach efforts, helping secure over **₹80,000** from multiple corporate and local partners. Maintained detailed financial records and supported resource mobilization worth **₹2 lakh+**, ensuring smooth event execution and effective budget management.

CERTIFICATIONS

Cybersecurity: Foundations of Cybersecurity - **Google, IBM, Cisco |** Cybersecurity Compliance, Frameworks & Regulation - **IBM** | Operating Systems: Administration & Security - **IBM**
AI / Python & NLP: Programming in Python - **Meta** | PyEdge: Practical Python & NLP - **VIPS**
Web & Frontend: Introduction to Frontend Development - **Meta** | HTML & CSS in Depth - **Meta** | Programming with JavaScript - **Meta**
Mobile & Tools: Mobile Application Development (Flutter & Dart) - **VIPS** | Version Control with Git & GitHub - **Meta**

NON-SCHOLASTIC ACHIEVEMENTS

Performing Arts: Certified in Drama and nukkad natak - multiple DoE (Department of Education) certificates, Trained guitarist with 10+ years of consistent practice.
Sports: Zonal medallist in Shot Put & Discus Throw, Zonal medallist in Skating, Represented school at State level in Badminton.